[date of submission]

Lucia Faravelli, Editor
Structural Control and Health Monitoring
Wiley

Dear Editor,

Submission of an original research article: “Blind loci and design criteria for integral load-monitoring channels on continuous girder bridges, with laboratory verification and an assessment on public field records”

We submit the above manuscript for consideration as a Research Article in *Structural Control and Health Monitoring*.

Influence-area bridge weigh-in-motion, introduced by Ojio and Yamada, estimates the gross weight of a vehicle from the area under one response record. That this area equals the response to a uniform unit load is classical, and so is the combination of several channels by generalised least squares. The manuscript states both as such and uses them to derive design criteria for continuous girders, on which positive and negative influence areas partly cancel.

We claim as new:

1. a closed-form catalogue of blind loci for deflection, strain, rotation, shear and reaction channels on two- and three-span and haunched girders, including the near-blind band that can contain strain gauges set back from the piers;

2. a strain indicator that needs no reference rigidity, a cancellation ratio that separates cancellation from restraint, and a normalisation of the deflection indicator that makes its threshold explicit;

3. a placement rule: optimal sensors lie at the extremes of the uniform-load response, for strain at the extremes of $c\,M_q/E\,I$;

4. an identifiability analysis under uncertain pier stiffness, in which the constant follows a Möbius law in the scaled pier stiffness, and which names the measurement that resolves it;

5. an evidence-based assessment, on a laboratory beam and on public moving-truck records from a five-span box girder, of what single- and multi-channel integral estimators can and cannot deliver.

The field assessment reports its results with 95 % confidence intervals and equivalence tests (per-channel tables in the Supplementary Material), and it reports negative results. The reference weights on these records vary so little that predicting the mean weight beats every estimator, so the records cannot test weight discrimination. Constants are consistent with axle-configuration invariance, but the test cannot resolve differences smaller than about 27 %. Generalised least squares with the measured noise covariance performed worse out of sample than the gauge average, because crossing residuals are much less correlated between gauges than vehicle-free noise; we present this as a design caution for zero-sum (offset-immune) estimation. An application to an in-service 55 + 90 + 55 m box girder shows that the constant of its side-span deflection gauge is not identifiable from static acceptance data, and predicts the measurements that would decide between the model calibrations.

We believe the paper fits the journal's interest in sensor placement, identifiability, signal processing and field measurement. The journal has recently published work on deflection influence lines of continuous beam bridges, which we cite and distinguish, as we do a recent absolute-area weighing method for continuous rigid-frame bridges.

Data availability. The field strain records are from the public NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0 licence). The laboratory data, the derived data behind every figure and table, and all analysis scripts with their console output are provided as Supplementary Material, as are the processed data of the in-service bridge.

The manuscript is original, has not been published, and is not under consideration by any other journal. Two other papers by the authors are under review at *Structures*; they address different problems and share no data, test rig or text with this manuscript. All authors have approved the submission. Conflicts of interest, funding and the use of generative AI are declared in the manuscript.

Yours sincerely,

Thai-Son Vu (corresponding author)
Faculty of Bridge and Road Engineering, Hanoi University of Civil Engineering

55 Giai Phong, Bach Mai ward, 11618 Ha Noi, Viet Nam
E-mail: sonvt2@huce.edu.vn

on behalf of Quoc-Bao Nguyen and Hung-Son Nguyen